

🎓 Education

University of Pennsylvania

Sep 2025 - Aug 2026

Visiting PhD Student

GRASP Lab · Advisor: Prof. Kostas Daniilidis

ShanghaiTech University

Sep 2021 - Jun 2027

PhD Student in Computer Science

Research interests: Visual Localization · 3D Reconstruction · Robotic Perception

Advisor: Prof. Laurent Kneip

ShanghaiTech University

Sep 2017 - Jun 2021

B.Eng. in Computer Science

🏛️ Selected Research

I Multi-view Geometry

A Linear N-Point Solver for Structure and Motion from Asynchronous Tracks

Oct 2025

ICCV 2025 Highlight (2.3%) · First author

- Established a unified geometry relating arbitrarily timestamped 2D observations, 3D points, and camera motion, and derived a linear N-point solver for joint structure and egomotion recovery.
- The solver takes point tracks from an upstream tracker and is implemented in C++ with Eigen and Ceres, supporting global-shutter, rolling-shutter, and event cameras.

An N-Point Linear Solver for Line and Motion Estimation with Event Cameras

Jun 2024

CVPR 2024 Oral (0.8%)

- Extended the 5-point event-camera solver to a singularity-free linear N-point solver that handles redundant line observations without polynomial root finding, improving accuracy, numerical stability, and speed.
- Fused multiple lines with geometry-aware residuals to recover full camera velocity and combined the solver with GC-RANSAC as the complete robust-estimation pipeline.

A 5-point minimal solver for event camera relative motion estimation

Oct 2023

ICCV 2023 Oral (1.8%) · Co-first author

- Established a geometric model linking event-camera line features, spatiotemporal observations, and camera motion, and derived a 5-point minimal solver that jointly estimates 3D line parameters and camera translation under known rotation.
- Fused observations from multiple lines to recover full translation, capturing more inliers and improving stability over spatiotemporal-plane clustering.

Event-IMU velocity estimation

Mar 2026 - Present

- Fused event flow with IMU for VIO velocity initialization; further analyzed flow-inertial observability.

I Learning-based Matching & Data Generation

Match-Any-Events: Zero-Shot Motion-Robust Feature Matching Across Wide Baselines for Event Cameras

Jun 2026

ECCV 2026

- Designed multi-timescale event feature fusion and a sparsity-aware attention block, implemented with a DINOv3-pretrained ViT for efficient semi-dense matching.
- Outperformed VGGT and MatchAnything by over 20% on zero-shot event-camera matching, with support for fast motion, wide-baseline matching, object pose estimation, and COLMAP reconstruction.

3DGS-based event simulator for event-camera network training

Oct 2024 - Jan 2025

- Reconstructed 3DGS scenes from ScanNet, augmented the original camera trajectories with synthesized variants, and rendered RGB videos; generated event streams from the rendered videos with a video-to-events network, then used FlowFormer and CoTracker to produce pseudo-ground-truth optical flow and point tracks.

| **Marker-based Localization & Visible-Light Communication**

Motion-Aware Optical Camera Communication with Event Cameras	Nov 2024
<i>IEEE Robotics and Automation Letters</i> , Presented at ICRA 2025 · First author	
➤ Designed dynamic visual markers and an event-camera detection, tracking, and decoding pipeline, building an end-to-end visible-light communication system for simultaneous localization and data transmission under camera motion. ➤ Achieved 1 cm localization accuracy with a 1% bit error rate under camera motion and up to 114 Kbps static throughput, with validation in a real-world AR scenario.	
Event-camera and LED-array localization and visible-light communication system	2024 - Present
➤ Redesigned LED codes and built a C++/ROS system with Eigen and Ceres for localization and 300 Kbps communication.	

| **Other Research**

Denoised internal models: a brain-inspired autoencoder against adversarial attacks	Oct 2022
<i>Machine Intelligence Research</i>	
➤ Designed a two-stage generative model combining input denoising and class-conditional reconstruction to improve the adversarial robustness of visual classifiers. ➤ Evaluated the model across 42 attacks, outperforming contemporary methods in overall robustness while retaining efficient inference.	

Reviewer: CVPR, NeurIPS, ECCV, BMVC, ICRA, IROS, IEEE TR-O, IEEE T-MECH, IEEE RA-L

 Experience

Changan Automobile (Chongqing, China)	Aug 2021
Intern at Autonomous Driving Group	
➤ Read internal documentation on autonomous-driving solutions and test scenarios, and attended discussions on underground-parking scenarios, user needs, and driver-monitoring anomalies.	
ShanghaiTech University	2020 - 2022
Teaching Assistant	
Machine Learning; Introduction to Programming; Computer Architecture	

 Honors and Awards

Outstanding Student (Top 10%)	2022, 2024, 2025
Merit Student (Top 5%)	2023
Outstanding Thesis (Top 5%)	Jul 2021
Second Prize, Shanghai, China Undergraduate Mathematical Contest in Modeling	2020
First Prize, National Olympiad in Informatics in Provinces (NOIP)	Nov 2015

 Skills

Languages:	C++, Python
Tools & Frameworks:	PyTorch, ROS, NumPy, Eigen, Ceres, CMake, Git, COLMAP, Pixi